“The Principle of Relativity”

K. Kroman

Fysisk Tidsskrift 14 (1915–16), pp. 1–30

Translated from the Danish

I.

It is characteristic of the recent period that several of the more rigorous sciences have reached such a stage of development that the researchers concerned have felt themselves prompted to examine more closely the reliability of the very foundation upon which their science rests. Thus the mathematicians have for a good many years been occupied with investigations concerning the foundation of geometry, a labour which, as is well known, has so far led to the not very gratifying result that even in the mathematical domain there is no longer agreement among men, since one circle of researchers maintains that the *Euclidean* basis continues to be the only right one, while another circle finds it more correct to set up as many as half a score of different and equally legitimate systems, of which *Euclid*’s is only one.

In the domain of physics too, relativity endeavours of various kinds have made themselves felt in recent years. *Newton* was, as is well known, the one who first clearly and exhaustively set up a general foundation for physics — as it was known in his day. But at that time it consisted essentially only of what we now call mechanical physics. Of the other main divisions of present-day physics there were as yet but faint traces. Both *Huygens*’ and *Newton*’s optics were still essentially purely mechanical. In our day there has come in particular an extensive theory of electricity, and with it there has arisen, among other things, the question: can the manifold electrical phenomena also be explained on the *Newton*ian basis, or does this now require additions, indeed perhaps transformations?

Owing to the imperfect knowledge we still have of the real nature of electricity, some time will surely yet pass before a fully grounded answer can be given to the comprehensive question just mentioned, and it is therefore not this problem that is to be treated in the following pages. It is rather a relativity problem of a somewhat narrower kind, namely the *Einstein*ian hypothesis for the explanation of the *Michelson* light-velocity experiment, that is to be made the object of a few critical remarks.

A. *Einstein* first presented his views in *Annalen der Physik*, 1905, and his treatise has been published, together with several others on the same subject, in 1913 as a separate little work with the title: H. A. *Lorentz*, A. *Einstein*, H. *Minkowski*: *Das Relativitätsprinzip*.

The *Einstein*ian hypothesis, which rests chiefly on the presupposition that it is impossible for us to make any difference between rest and uniform rectilinear motion, at once aroused considerable attention and met with no slight adherence; there appeared a fairly luxuriant literature which treated the problem and sought to carry it further. There were, however, also many who took a rather sceptical attitude towards his thoughts, and among others both H. *Poincaré* (in a lecture delivered in Berlin: *Die neue Mechanik*, issued in a 2nd edition in 1913) and H. A. *Lorentz* (in *Das Relativitätsprinzip*, 3 lectures, 1914) have expressed particular doubt as to whether the way out thus indicated can be reconciled with the fundamental conditions of human knowledge.

It seems to me that the eminent researchers named are right in the doubt they have indicated, and it is therefore especially the *Einstein*ian hypothesis, seen from the epistemological side, that shall be sought to be examined more closely in what follows.

II.

The first occasion of the relativity hypothesis was certain phenomena of light. The theory of light seems on the whole to be one of the physicists' problem children. *Huygens* sets up his wave theory; but *Newton* points out that wave motions can bend round corners, whereas light rays apparently never do so. Neither could the wave theory manage polarisation, since for the time being only longitudinal vibrations were conceived. *Newton* therefore sets up his emission theory, which wins general acceptance and holds its ground until it appears that it too leads to contradictions. The difficulties in the wave theory have meanwhile been removed, among other things by exchanging the longitudinal vibrations for transverse ones. Unfortunately, however, vibrations must be vibrations of something or in something, and one was thereby led to set up that remarkable "medium" which is called the ether. It was the elasticity of the ether that was to maintain the rapid vibrations. One therefore had to ascribe to it an incredibly great coefficient of elasticity; indeed, since the vibrations were transverse, one had rather to conceive it as a kind of solid body, while at the same time one found that it apparently offered the world-globes not the very slightest resistance. By mathematical means the theory was meanwhile worked out more and more finely, so that it could by degrees give the most excellent solution of a multitude of particular problems, and the many elegant results cast in part a veil over the brittle foundation. An essential advance took place besides in that *Maxwell* exchanged the elastic vibrations for electromagnetic ones. The transverse vibrations thereby became explicable, and the ether lost something of its strangeness. But an enigmatic medium it nevertheless continued to be under the *Maxwell*ian presuppositions too, and in its last form as well the theory was soon to lead to apparently insuperable difficulties.

It had been possible to find the velocity of light in the world-space, in air, in water and in a multitude of other bodies. One believed further that one had found a quite simple explanation of the astronomical phenomenon of aberration. If a telescope is thought of as pointing from the earth straight at the sun, a light ray which strikes exactly the centre of the objective will not strike the centre of the eyepiece; for while the light traverses the length of the telescope, earth and telescope have moved $^1/_{10\,000}$ of this length to the side, so that the ray will form an angle with the telescope's axis of about 20.5 seconds of arc. Natural and obvious as this explanation may seem to be, it nevertheless contains a definite positive presupposition, namely this, that the light ray retains its direction in space while the telescope moves to the side, and that the ether, in which the light motion takes place, thus rests in space and is not dragged along by the earth's motion. It is the hypothesis of *the resting ether* that is thereby set up. It is, as one sees, in good agreement with the circumstance that the globes seem to revolve without suffering the slightest resistance from the ether.

Airy filled a telescope with water and yet obtained through it the usual aberration constant; this experiment might seem fateful for the usual theory of aberration, since light takes $^4/_3$ times as long in water as in air. But it is of course possible, indeed probable, that light cannot move through water which has a lateral velocity of 30 km per second without taking along a part of the water's velocity, and if it took, for example, $^1/_4$ of this, all was of course in order. Now both *Fizeau* and *Michelson* have shown by experiment that light, in going through water with a velocity in the direction of the ray, receives a part of the water's velocity into the bargain. *Fresnel* had, on certain presuppositions about the ether, predicted that this increment must be $\frac{n^2-1}{n^2}$ times the water's velocity, n being the water's refractive index. On more recent presuppositions one has come to the same result, and the experiments of the two researchers fully confirmed this.

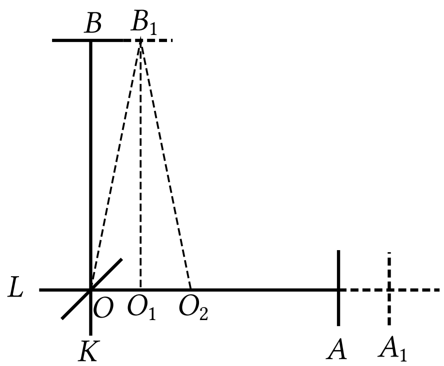
The decisive difficulty for current physics therefore first sets in with A. *Michelson*'s famous experiment. This has already been described in detail earlier here in this journal[*)].

*) Vol. 10, p. 257.

It will therefore be sufficient to recall its main points briefly.

III.

Michelson, who was famous for his fine interference experiments, surely expected by such an experiment to be able to furnish an indoor proof of the earth's translation, just as *Foucault* by his pendulum and his gyroscope had furnished an indoor proof of its rotation. He wished, in other words, to try whether one could not, by the aid of the displacement of the interference fringes, trace a difference in the time light took to run a certain relative stretch there and back, when the path fell the one time in the direction of the earth's translation and the other time stood perpendicular to it, Since the earth's translation velocity is about 30 km per second, thus about a ten-thousandth of the velocity of light in the world-space and in air, this did not seem to him impossible. He sent out then parallel light rays from L, which by the aid of a lightly silvered plane mirror at O divided, so that some went the one and others the other of the two paths there and back, whereupon they united again at O and went to the telescope at K. If the relative paths were equally long, the more absolute ones would of course become different, and the reunited rays would then have to show a displacement of the interference fringes. His whole apparatus swam upon mercury and could be turned without coming into appreciable disorder, so that now the one and now the other set of rays went the longer of the two paths.



If the earth's motion has the direction OA, and if $OA = OB = l$, a light ray would — he thought —, if the earth stood still, take the time $\frac{2l}{c}$ to go there and back between O and A or B, c being the velocity of light. But if the earth itself has the velocity $v = \frac{c}{10\,000}$, the ray which goes to A (which at the same time comes to A_1) must take for the outward path the time $t_1 = \frac{l + vt_1}{c} = \frac{l}{c - v}$ and for the return path the time $t_2 = \frac{l - vt_2}{c} = \frac{l}{c + v}$, thus in all the time $t = \frac{2l}{c} \cdot \frac{1}{1 - v^2/c^2}$.

The ray which goes to B will, on account of the earth's motion, traverse the path OB_1O_2 and must therefore take for the outward path the time $t_3 = \frac{\sqrt{l^2 + v^2 t_3^2}}{c}$ and for the homeward path the same time, thus in all the time $t' = \frac{2l}{c} \frac{1}{\sqrt{1 - v^2/c^2}}$.

The path difference for the two rays thus becomes, to a close approximation,

$$F = 2l \left[\frac{1}{1 - v^2/c^2} - \frac{1}{\sqrt{1 - v^2/c^2}} \right] = 2l \left[1 + \frac{v^2}{c^2} - \left(1 + \frac{v^2}{2c^2} \right) \right] = l \frac{v^2}{c^2} = l \cdot 10^{-8}.$$

At *Michelson*'s first experiment*) $l = 120$ cm, and the light he used had the wavelength $\lambda = 60 \cdot 10^{-6}$ cm. In a later experiment, which he carried out together with *Morley*, l had by the aid of mirrors been increased to 1200 cm. In the first case one obtains $F = \frac{12 \cdot 10^{-7}}{60 \cdot 10^{-6}} \lambda = 0.02\lambda$, in the last $F = \frac{12 \cdot 10^{-6}}{60 \cdot 10^{-6}} \lambda = 0.2\lambda$. One had then to expect a displacement of the interference fringes of the magnitude of 0.02 and 0.2 fringe-breadths respectively, now to the one and now to the other side, according to how the apparatus was oriented. Even the smallest of these displacements would according to *Michelson* have been observable in the telescope used, which was furnished with an ocular micrometer. But not even in the last case was any change with certainty seen in the position of the fringes.

This negative result was undeniably a surprise, a riddle which brought contradiction into our physical knowledge. It then became the task to get this contradiction removed again.

The most obvious way out was both attempted and barred by *Michelson* himself. He conceived the possibility that the earth at the moment of the experiment had no real translation either, the whole solar system going at the same time in the opposite direction with a velocity of 30 km. He therefore repeated the experiment after the direction of translation had become an essentially different one. But the result remained as before.

The next nearest way out would be that of declaring the experiment insufficiently perfect. Where it is only a matter of a path difference of a fraction of a light wavelength, obviously even the smallest inaccuracies can destroy the whole thing. Even the very slightest difference of temperature could of course alter the mutual distance of the mirrors sufficiently. To this it may however be answered that the distinctness of the interference fringes and their remaining in place showed decisively that the apparatus was otherwise in order, and that it was expressly the path difference that it refused to show any sign of.

*)According to *Wüllner*: *Experimentalphysik*, 5. Aufl. IV, 607. The numerical magnitudes are given extremely differently in the various accounts. They play, however, no essential part in the present connection.

We must then seek new ways out, and the methodically right course is surely that we first and foremost turn to the current theory of light itself and ask: is it then entirely in order? Was one right in expecting the fringe displacements mentioned?

Here it seems to me that there lies a widely extended field for closer investigations, a field which surely cannot already be so thoroughly ploughed that one can give up all searching here and turn to all manner of more remote possible ways out. As long as interference phenomena have been known, they have of course caused research surprises which it was only gradually possible to master. Do we know with sufficient clarity how the luminous bodies go about producing and emitting the remarkable light vibrations? It is presumably the general view that light in the world-space and in the atmosphere is constantly emitted and propagates itself with the constant velocity $c = 300\,000$ km per second, independently of the light source's rest or motion. But what secure grounds has this assumption? Might it not be that it has formed itself rather involuntarily, merely because the possible increments or deductions that could come into question here seemed so insignificant? It has been wished to explain *Michelson*'s result by the assumption that light constantly took along the light source's velocity as an increment. Objections have been raised against such a view; but against these objections too objections have been raised. A gentler assumption, however, was that light in the first moments took along the light source's motion, but quickly passed over to the usual velocity. That the current theory should be so firmly formed that there were no room at all for one or the other of such alterations is surely a hasty belief. Neither is the behaviour of the ether presumably as yet so firmly established that there should be no room here for one nuance or another which would make a solution of the riddle possible. A number of experiments have of course followed *Michelson*'s with closely corresponding negative results. But the most essential of them have all lain in the domain of light phenomena, and it therefore seems fairly reasonable that it ought also to be there that one looks first and foremost for the redeeming way out.

It looks, however, as if the physicists had already grown weary of searching here. The first attempt at a solution which won a wider adherence had a far broader basis. It came from H. A. *Lorentz* and was distinguished in any case by being after all a really *physical* attempt at a solution.

Lorentz thought that it would meet with insuperable difficulties to conceive the ether as carried along by bodies, and he felt himself likewise convinced that light constantly has the velocity c through the ether and in air. He therefore assumed that *Michelson*'s negative result must rest upon this, that every translationally moved body suffers a contraction in the direction of the motion just great enough that, among other things, *Michelson*'s result thereby becomes explicable[*]. If the translation velocity is v, the body's extension in the direction of translation becomes, according to the hypothesis,

$$a = \frac{1}{\sqrt{1 - v^2/c^2}}$$

times as small as it was at rest. During *Michelson*'s experiment his apparatus had of course the earth's translation velocity, v. It has thus in this direction become a times as small. But precisely thereby the two light paths have constantly become equally long.

Lorentz remarks that this hypothesis must indeed at the first moment awaken some wonder. But on closer consideration it becomes after all quite natural and obvious. The molecular forces must surely in the last resort be conditioned by the ether, and it can then be no surprise that they can suffer the assumed slight change in translation. For it is only an exceedingly slight change that is in question. The magnitude a is only quite imperceptibly different from 1. The earth's orbital velocity is about 1800 times as great as that of one of our express trains, and the earth's diameter is about 12 800 km. And yet its contraction would amount only to 6.4 cm. That no one has hitherto noticed that things suffered such a contraction can therefore cause no wonder.

This is the *Lorentz*ian explanation of *Michelson*'s negative results. It is distinguished, as has been said, by being a really physical hypothesis, which in its way really explains what was to be explained. Several — among others H. *Poincaré* — have, however, found it extremely difficult to assume such a physical contraction, which should be the same for the most different bodies, provided only these had the same velocity, and, as we shall see, the *Einstein*ian train of thought is then also directed at explaining the results in question without the introduction of any real physical contraction, indeed, in Einstein's own view, without introducing any properly new hypothesis at all.

IV.

The first impulse to his peculiar attempt at a solution *Einstein* surely received from the *Lorentz*ian contraction hypothesis; but for the rest he has gone his own quite particular ways. We shall begin by deriving his famous formulae without as yet building upon his particular presuppositions. These formulae can namely be obtained from several different starting points, and moreover by a quite elementary route, and if we proceed in this way while for the present keeping to quite ordinary presuppositions, we shall in part

[*] The same way out was proposed at about the same time by *Fitzgerald*.

perhaps succeed in getting removed the half-mystical gleam which the formulae have often enough acquired, and in part the subsequent transition to the particular *Einstein*ian presuppositions will most clearly let these come forward in their full peculiarity. In spite of this we shall not need to remove ourselves very far from his own presentation.

We think of ourselves, then, as having in space two rectangular coordinate systems, $Oxyz$ and $O'x'y'z'$. At a certain moment they coincide completely; but the latter has, in relation to the former, a translation in the direction of the X-axis with the constant velocity v. We can think of a corporeal world attached to each of the two systems, and assume an observer, A, in the first and an observer, B, in the second, each with his collaborators furnished with mechanically perfect clocks and measuring rods. Each of them believes his own system to be at rest and the other's to be in motion, and they both assume that light in the world-space and in air has the constant velocity c independently of the light source's rest or motion. It is now a matter of finding out what, in accordance with their different presuppositions, they must think about each other.

We may begin with A. He sets up his clocks at different places in his system and procures for them the same rate and setting by sending out light signals which are thrown back to the starting point. Let the origins of both systems reckon the time from the moment of their coincidence, and let us for convenience assume that all operations take their beginning at this point of time. A thus sends at the time 0 a light signal from O to the place L at the distance l. On the signal's arrival at L the clock there is thus, according to his presupposition, to be set at $t_1 = l/c$, and on the signal's return to O the clock there is to show $t_2 = 2l/c$. In this way A seeks to procure for all his erected clocks the same rate and reading.

B, who believes *his* system to be at rest, goes about it in quite the same way. He sends, for example, at the time $t' = 0$ from O' a signal along the Y'-axis to a clock at the distance l. This is set on the signal's arrival at $t'_1 = l/c$, and the clock at O' is set on the signal's return at $t'_2 = 2\,l/c$.

This procedure appears, however, to A, who believes B's system to be in motion, as quite incorrect. He finds that the light must have taken for the outward path the time $t_1 = \dfrac{\sqrt{l^2 + v^2 t_1^2}}{c}$ and for the return path the same time, thus in all the time $t = \dfrac{2l}{c}\dfrac{1}{\sqrt{1 - v^2/c^2}}$. But if the magnitude

$$\frac{1}{\sqrt{1 - v^2/c^2}} = a, \qquad 1.$$

is set, A thus finds that B's time-indications must so far be a times too small and his time-units (seconds) a times too great. And for B's light signals along the Z'-axis he comes to the same result.

Along the X'-axis, however, the matter becomes more composite. B goes about it here as before. But A finds that if B sends a signal from O' to a place on the X'-axis at the distance l, it must take for the outward path the time $t_1 = \dfrac{l + vt_1}{c} = \dfrac{l}{c-v}$ and for the homeward path the time $t_2 = \dfrac{l - vt_2}{c} = \dfrac{l}{c+v}$, thus in all the time $t = \dfrac{2l}{c}\dfrac{1}{1 - v^2/c^2} = \dfrac{2l}{c} \cdot a^2$, and thus a times as long a time as under the corresponding conditions along the other axes.

If such a contradiction really took place, B must however — A thinks — quickly have discovered it. For it would then have been impossible for him to get his clock at O' regulated. It can therefore *not* take place. But this can only be conceived possible if all X'-distances l have in reality had only the length l/a. B's system must thus, on account of its motion, have contracted, so that it itself and all measuring rods and other bodies in it which have taken part in the motion have become a times as short in the X'-direction as they were at rest. If the measuring rods too are shortened, one understands that B has noticed nothing and that he has continued to measure the X'-lengths $\dfrac{l}{a}$ as l. But if one assumes this contraction, all B's signals there and back will thus have taken only a times as long a time as calculated by B, and one then gets everywhere in B's system (at rest in the system) clocks which go a times too slowly in relation to A's, thus so far an ordinary A-time and an ordinary B-time.

This holds, however, only of the clocks' rate.

For with respect to their setting there is for A still something amiss. A found that the signal in the X'-direction must for the outward path take the A-time $\dfrac{l}{c-v}$. On account of the contraction this must be corrected to the A-time $\dfrac{l}{a(c-v)}$, thus to the B-time $\dfrac{l}{a^2(c-v)}$. But B had the clock at his distance l set at the number $\dfrac{l}{c}$. The clock at the B-distance l from O' in the X'-direction is thus set so that it, according to A, gets a deficit of clock-reading in B-time of

$$u = \frac{l}{a^2(c-v)} - \frac{l}{c} = \frac{l(c^2 - v^2)}{c^2(c-v)} - \frac{lc}{c^2} = \frac{v}{c^2}\, l. \qquad 2.$$

A has thus further found that B's clocks have a deficit of reading proportional to their X'-distances from O'. If this distance is negative, the deficit is negative. At O' it is 0. In system B there is thus for A not only B-time, but in addition *local time*, much as we here on earth have not only mean time and sidereal time, but further Greenwich time and local time or zone time.

Let there now occur some momentary and punctual phenomenon, F, which by A must be referred to the time t and the place x, y, z and by B to the time t' and the place x', y', z', and let us determine the relations between the 8 magnitudes named.

According to the foregoing we can at once set

$$y = y' \qquad\qquad z = z' \tag{3. 4.}$$

Further it is easily seen that we must set

$$t = a\left(t' + \frac{v}{c^2}\,x'\right). \tag{5.}$$

For if t' is the B-time at the place x', y', z', the parenthesis gives us the B-time at O', and the A-time is precisely a times as great as this.

In order to get the relation between x and x' we set A's coordinate $x = m+n$, where m is the distance from O to O' at the A-time t, thus $= vt$ or according to 5 $= va\left(t' + \frac{v}{c^2}\,x'\right)$, while n is the X'-distance from O' to the phenomenon, thus x' B-metres or $\frac{x'}{a}$ A-metres. We thus obtain, since $\frac{1}{a^2} = \frac{c^2 - v^2}{c^2}$,

$$x = a\left(x'\,\frac{v^2 + c^2 - v^2}{c^2} + vt'\right) = a\,(x' + vt'). \tag{6.}$$

From 5 and 6 t' is obtained, since one has

$$t' = \frac{t}{a} - \frac{v}{c^2}\,x' = \frac{t}{a} - \frac{v}{c^2}\left(\frac{x}{a} - vt'\right); \quad t'\,\frac{c^2 - v^2}{c^2} = \frac{1}{a}\left(t - \frac{v}{c^2}\,x\right)$$

thus

$$t' = a\left(t - \frac{v}{c^2}\,x\right). \tag{7.}$$

And from 6 and 7 x' is obtained, since

$$x' = \frac{x}{a} - vt' = \frac{x}{a} - avt + a\,\frac{v^2}{c^2}\,x = a\left(x\,\frac{c^2 - v^2 + v^2}{c^2} - vt\right);$$

$$x' = a\,(x - vt). \tag{8.}$$

In all one thus obtains

$$\left.\begin{aligned} t &= a\left(t' + \frac{v}{c^2}\,x'\right) & t' &= a\left(t - \frac{v}{c^2}\,x\right) \\ x &= a\,(x' + vt') & x' &= a\,(x - vt) \\ y &= y' & y' &= y \\ z &= z' & z' &= z \end{aligned}\right\} \qquad 9.$$

As is seen, the one set of formulae is obtained from the other merely by letting v change sign and changing the unmarked letters into marked ones and conversely. This is an expression of the fact that B must think exactly the same about A as A thinks about B, apart from the direction of the velocity, and this we could have predicted, since B of course believed precisely the same about A as A about B, with the difference named.

From these fundamental formulae a multitude of others can of course be derived. We shall here merely add further the formulae for the constant velocity along each of the coordinate axes. If the motion begins at O at the time 0, one has of course $h_x = \frac{x}{t}$, $h_y = \frac{y}{t}$, $h_z = \frac{z}{t}$ and of course $h'_x = \frac{x'}{t'}$, $h'_y = \frac{y'}{t'}$, $h_z = \frac{z'}{t'}$. But then from 9 one obtains

$$h_x = \frac{a(x' + vt')}{a\left(t' + \frac{v}{c^2}\,x'\right)} = \frac{h'_x + v}{1 + \frac{v}{c^2}\,h'_x}$$

and by the same route on the whole

$$\left.\begin{aligned} h_x &= \frac{h'_x + v}{1 + \frac{v}{c^2}\,h'_x} & h'_x &= \frac{h_x - v}{1 - \frac{v}{c^2}\,h_x} \\ h_y &= \frac{h'_y}{a\left(1 + \frac{v}{c^2}\,h'_x\right)} & h'_y &= \frac{h_y}{a\left(1 - \frac{v}{c^2}\,h_x\right)} \\ h_z &= \frac{h'_z}{a\left(1 + \frac{v}{c^2}\,h'_x\right)} & h'_z &= \frac{h_z}{a\left(1 - \frac{v}{c^2}\,h_x\right)} \end{aligned}\right\} \qquad 10.$$

By an easy differentiation the same expressions are obtained for variable velocities. All the formulae together form what the mathematicians call “an invariant transformation”.

V.

All this is however for the present merely a mathematical fantasy. We shall now try to bring it into relation with reality. First we consider it in quite the usual way.

A believes his system rests. Let us, in order to get a definite starting point, *for the present* presuppose that he is right; no one has after all thought of abolishing the concept of rest outright. Further, both A and B entertain the usual view that light, independently of the light source's rest or motion, has in the world-space and in air the constant velocity c. Let us also for the present presuppose that this is correct. Then A regulates his clocks in the manner described. To that we must grant him a complete right. Further he finds that if B's system has the constant translation velocity v, B cannot rightly regulate his clocks as he does. They must thereby come to go too slowly, and even if, in order to make B's behaviour intelligible, one assumes the contraction mentioned, they must further get local time. These assumptions of A's we must likewise declare justified on the basis of his presuppositions, which we for the present share. On the other hand, on the basis of these presuppositions we cannot find B's proceeding justified. We must maintain with A that B is in error. If A rests, B cannot rest, when there is the velocity difference v between them. And if we place ourselves at B's standpoint and believe him to be at rest, we can no longer believe A to be at rest, but must then declare A's whole proceeding unjustified. Possibly they are both wrong. But the decisive thing is that we cannot at one and the same time give them both right. Their starting points cannot be reconciled with each other. Their results stand in contradiction with each other.

Einstein's view is however quite another. He too builds upon the above-mentioned presupposition about the velocity of light. But he adds to it the *new* presupposition that the optical (electrodynamic) phenomena "like the mechanical ones" take place in the same way, whether they are referred to a system at rest or to a system with uniform translation. It is this presupposition that he calls "the principle of relativity". By the aid of it he will explain *Michelson*'s negative results, and in these results themselves he finds a guarantee of the presupposition's correctness. His view of A and B therefore becomes this:

A can (according to the relativity presupposition) never obtain certainty that his system rests. But he can obtain certainty that it has no acceleration, and if he has assured himself of this, he also has (according to the relativity presupposition) the right to measure his lengths and regulate his clocks as he did. He can also obtain certainty that B has the constant relative velocity v, and according to the presupposition of the constant velocity of light he can then also rightly judge of B's clocks and whole system as he did. But B is on the other hand also right in his proceeding. For whether he rests or not, provided only he has no acceleration, he incurs (according to the relativity presupposition) no error in reckoning himself at rest.

Here, then, there are *two truths.* A is right; for he has no acceleration. And B is also right; for he has no acceleration either. But this obviously will not do. There is in logic a fundamental proposition which says: A is not not-A; contradictions (in the world of things) do not exist, and contradictions (in the world of thought) are not tolerated. Truth is not double. Truth is only one. This proposition needs no defence. Every science is built upon it. *Einstein*'s theory, on the contrary, stands in conflict with it. Therefore it must be rejected.

Of course it has not been clear to *Einstein* that his doctrine was in conflict with the principle of contradiction. Then he would surely have given it up. He has believed that the conflict was only apparent. He himself emphasises that it looks as if there were a contradiction between his two presuppositions; but — he adds — this is not so. In this, however, he is wrong. The two presuppositions do not admit of being united. One needs only to ask: what should B have done if he too had himself believed his system to be in motion (without acceleration)? Should he have maintained his results? Then the constant velocity of light would of course have been given up. Should he have reckoned as A did? Then "the principle of relativity" would of course have been given up. The two presuppositions do not admit of being reconciled, simply because the one says shortly and plainly: light's velocity is c, while the other says: light's velocity is sometimes $c + v$ and sometimes $c - v$ and sometimes something in between. This difference can be concealed by one's using different clocks. But it is not thereby out of the world. In the domain of time either logic does not permit us to believe in two truths. We can perfectly well believe in both sidereal time and mean time. But we find it necessary to keep to one of them at a time when we want to compare velocities.

Einstein has surely believed that he could appeal to another logical proposition in his defence. For logic says further: *Diversi respectus tollunt omnem contradictionem.* If a matter is seen from two sides, there is no contradiction in different judgements being passed about it. A musician can, for example, be *great* as a virtuoso but *small* as a composer. And in the same way A's proceeding may be right from A's point of view and B's from B's point of view.

This way out, however, does not hold either. When the two judges of the musician speak fully out, it will appear that we have here *two different judgements about two different things.* In that there is no contradiction. A third party present may quite well hold both judgements to be right and both judges to be fully competent. But suppose they had both spoken of the musician as a virtuoso. Then the contradiction would remain standing for the present, and the third party would get no order into his consciousness until he had learned, for example, that the one of the judges was unconditionally competent and the other unconditionally impossible. But *Einstein*'s case is precisely of the latter kind; we do not get two judgements about two things, but two judgements about one thing. B's system is contracted, says A. No, says B. B's clocks go too slowly and have local time, says A. No, says B. What is the unfortunate third party now to think? *Einstein* has really no other answer than: both! But with that we have the two truths again.

What is the time? a Bornholmer asks a tourist from London. It is twelve, answers the man. That agrees, says the Bornholmer; for my clock is just one, and the mean sun comes into the meridian here just an hour before it comes into the meridian for Greenwich. — With that the contradiction is out of the world. Here too we had merely two judgements about *two* things. But in this way A and B will never be able to come to agreement. It looks as if *Einstein* had presupposed that the physicist was only A or B. But the physicist is unfortunately the third party, who is to procure unity between A and B, and this unity is impossible. *Lorentz* has hit the theory's weakness right in the centre when he somewhere hints that it will be bad if A and B ever come to speak with each other[*)].

Einstein will perhaps appeal to the fact that he has really done nothing but describe how the physicists here in the world actually go about it. Like system B, the earth has of course, according to their view, a translation v. And yet they go about it like B and regulate their clocks by light signals quite as he does. They must thus also get B-time and local time of the same kind as B's, and precisely thereby the *Michelson* result becomes explicable.

To this it is however to be remarked that this contention is untenable. The great majority of our physicists are undoubtedly more or less convinced that it must after all surely play a certain part whether the light goes in the direction of the earth's translation or against it, and it is only because they know at the same time that the exceedingly slight difference which can hereby arise will in the great majority of cases be far exceeded by the various imperfections with which almost all our measurements are encumbered that they disregard it. But in return they are then also perfectly clear that it is only a greater or smaller approximation, and not the absolute truth, that they have reached. Besides, no physicist of course contents himself with the light signals mentioned; every physicist constantly has in addition other routes by which he seeks to win the most reliable and exact result that can be attained at all. And finally the *Michelson* result was of course

[*)]Lorentz: Das Relativitätsprinzip, 22.—23.

conditioned neither by any clock rate nor by any clock setting. It admits of explanation if we assume the *Lorentz*ian contraction hypothesis. But with *Einstein* the contraction is of course merely a *conjecture* in A's consciousness, called forth by his endeavour to get coherence into B's' behaviour. Taken as a reality it stands with *Einstein* quite *without cause.*

Both *Einstein* and most of his adherents have with great zeal emphasised something which they call *Newton*'s "principle of relativity", and have often expressed themselves as if it were only a *Newton*ian thought that they had developed further. Here too, however, a misunderstanding makes itself felt. It is in reality something brand new that *Einstein* has uttered, different from everything one finds in *Newton*. *Newton* was not in the least degree a relativist in the *Einstein*ian sense of the word. He was, as he has himself set it forth in the introduction to the *Principia*, convinced that there was something which had to be called absolute rest and absolute motion, and something which was only relative rest or motion, and he was likewise convinced that rest and motion were on the whole different states. This, he thought, our thought compelled us to assume. In rational mechanics we have of course also constantly been able, clearly and without contradiction, to make a difference between rest and relative and absolute motion. We have there set up for these matters an extensive theory which hardly anyone will raise objection to. In the domain of thought, therefore, all is in order. But in the face of *the given real case* uncertainty can arise about what we really have before us, since we, as *Newton* says, here lack a secure coordinate system. Often we can by closer investigation reach a decision; but — he adds — sometimes it can be "hard". That it should ever be simply and plainly impossible he has not committed himself to asserting. He has thus stated that there was here a certain — perhaps provisional, perhaps lasting — limitation in our insight. But if anyone had asserted to him that rest and uniform rectilinear motion were physically one and the same, he would quite certainly have protested, and evidently with right. The mathematical concept "the invariant transformation" is presumably a particularly valuable mathematical concept. But one must take care not to misuse it physically in exaggerated abstractness. When C and D play ball in the cabin of a ship — it is said —, everything will go on in the same way independently of whether the ship lies still or has uniform translation. This is only a half-truth. C or D or the spectators will perhaps not discover any difference. But the physicist, who looks more searchingly at the matter, will hardly find the assertion exact or exhaustive. He will involuntarily conjecture that since a resting and a uniformly translationally moved body cannot after all be apprehended by thought as completely one and the same, there must surely also be some difference in what happens in the two cases. And it will not be difficult for him to find out that if the ball is struck forward from C to D and back from D to C, then, if the ship moves, it must go a longer way the one time than the other, and if the time for both motions is the same, then the ball must necessarily have been acted upon in different ways by C and D, and one then also easily finds that to the arm's relative motion the ship's motion is constantly added, now as an increment and now as a deduction. But from this alone it is seen sufficiently clearly that the event in its details is different in the two cases, and that it is only the peculiar interplay of these different details that makes the final result the same for the spectators. Here, then, there is both difference and likeness; but the characteristic thing in the whole matter is this, that we can everywhere find *ratio sufficiens* for what happens, the one time as well as the other. The demand of thought, the principle of causality, is under each of the two conditions unambiguously observed. The two events come to look alike precisely because the individual members were different from the one time to the other, precisely because the ball each time *took along the starting point's velocity as an increment.* The whole matter is therefore here *rational.* With *Einstein*, on the contrary, the ball, i.e. the light, does *not* take along the starting point's velocity as an increment, as he himself expressly emphasises. And yet the phenomenon is to become the same in the two cases. This is *irrational.* This is again the two truths, which he believes he can transform into one merely by veiling them with a couple of formulae that contradict each other to just the right degree.

It is true that with *Newton*, and in ordinary physics generally, there are certain formulae which can remain the same in the resting and in the uniformly rectilinearly moved system. But from that one cannot yet conclude that there is identity between the two cases. A flying bird draws a little wagon of the mass m forward on the deck of a resting ship. The wagon begins with velocity 0 at the time 0. It has only inertial resistance and gets the acceleration γ. The bird must thus pull with the force $m\gamma$, and when the velocity has become γt and the way $\frac{1}{2}\gamma t^2$, the bird has applied the force $m\gamma$ through the way $\frac{1}{2}\gamma t^2$, thus the energy $\frac{1}{2}m\gamma^2 t^2$.

If the ship goes in the same direction as the wagon with the constant velocity v, the *ship* phenomenon may quite well become entirely the same, both in respect of the wagon's velocity, its acceleration and the action of force upon it. But the bird must now deliver the force $m\gamma$ through the path-length $\frac{1}{2}\gamma t^2 + vt$, thus the quantity of energy $m\gamma vt$ more than before. It is, among other things, only because we so often in our calculations deal freely with the force without interesting ourselves particularly in the *source of the force* that everything can come to look so alike. Anyone who stands outside both situations will also, with sufficient care, find them different from each other.

Einstein has thus not solved the riddle behind the *Michelson* phenomenon. He has merely moved it over into a couple of formulae, which he offers us as a kind of magic remedy against it. But he has overlooked that the medicine here is worse than the disease, that there is also a well-known formula which runs: the patient died, to be sure; but the fever left him! If there were a yawning abyss between A and B, then they might possibly live on happily, each with his own view. But such an abyss there is not, as we shall see. They must therefore try to come to agreement, just as the physicist must try to find a single sovereign truth. But towards the solution of that task *Einstein* has not furnished the slightest guidance. He has not even seen that there was a task here.

VI.

Let us consider a couple of *the theory's applications* and see whether they should bring us into conflict with the view here maintained[*)].

In the first place, then, we find a couple of simple consequences of the mutually contradictory starting points.

There lies a rod on system B's X'-axis from $x_1' = 0$ to $x_2' = l$. It thus has for B the length l and is for him at rest, while for A it is in motion. From formulae 9 we learn what length A must ascribe to it. One has

$$x_2' - x_1' = l = a(x_2 - vt) - a(x_1 - vt) = a(x_2 - x_1),$$

thus

$$x_2 - x_1 = \frac{l}{a}.$$

For A it is thus a times as short as for B.

If such a rod lay in the same way in A's system, we should get

$$x_2 - x_1 = l = a\,(x_2{}' + vt') - a(x_1' + vt') = a(x_2' - x_1'),$$

thus

$$x_2' - x_1' = \frac{l}{a}.$$

B thus in the same way finds A's rod contracted.

[*)] A mathematical formula can be true or false according as it is put in connection with the one basis or the other. We here look at the *Einstein*ian formulae from the *Einstein*ian basis alone.

We think at $t = t' = 0$ of a sphere with centre at O' and radius R. It thus has for B the equation

$$x'^2 + y'^2 + z'^2 = R^2.$$

But according to 9 one obtains at $t = t' = 0$ the relations $x' = ax$, $y' = y$, $z' = z$, thus

$$a^2x^2 + y^2 + z^2 = R^2.$$

For A the sphere has thus become an ellipsoid of revolution with the semi-axes $\frac{R}{a}$, R, R. It too has thus quite correctly received the contraction, in the direction of the motion, of the body assumed to be in motion. If it lay in A's system (which it in fact does at $t = 0$, since the systems coincide at $t = t' = 0$), A would, according to the formulae, apprehend it as a sphere and B find it contracted. Even this is of course a little strange. But the formulae thus do their duty. Purely mathematically there is nothing amiss. If, on the other hand, we think of A and B as equipped with eyes so perfect that they could *see* whether there was contraction or not, we stand again in the midst of the contradiction.

Let us take yet another example of the formulae yielding what they are contrived to yield.

From a light source at O there goes out at $t = 0$ a momentary flash of light. In virtue of the constant velocity of light c it will spread as a growing spherical shell which at the time t has the radius ct and is thus represented by the equation

$$x^2 + y^2 + z^2 - c^2t^2 = 0.$$

Thus it will present itself for A. How does B find it? Since $y = y'$ and $z = z'$, we need only seek the value of $x^2 - c^2t^2$. But

$$\begin{aligned} x^2 - c^2t^2 &= a^2(x'^2 + v^2t'^2 + 2x'vt') - c^2a^2\left(t'^2 + \tfrac{v^2}{c^4}x'^2 + 2t'\tfrac{v}{c^2}x'\right) \\ &= x'^2a^2\left(1 - \frac{v^2}{c^2}\right) - c^2t'^2a^2\left(1 - \frac{v^2}{c^2}\right) = x'^2 - c^2t'^2, \end{aligned}$$

thus

$$x'^2 + y'^2 + z'^2 - c^2t'^2 = 0.$$

The phenomenon thus becomes the same for B as for A. In B's system too "the light has thus had" the constant velocity c.

Some reader or other will perhaps exclaim in impatience: but this can surely never have come about naturally! No, wholly naturally it certainly has not. Thought cannot grasp that a point of light should be able to be in two places at the same time. But the impatience is out of place. For one then forgets that it is of course merely the mutually contradictory formulae that conjure the marvel forth. If A and B had the eyes spoken of before, they would quite certainly, by using them, get another result. But the formulae have thus here too done what they were contrived for. They have apparently taught us that light can at one and the same time have the velocity c both in a resting and in a moved coordinate system. If one is satisfied with the formulae, one has thus here obtained a "solution" of the *Michelson* riddle itself.

It will perhaps be objected: but these formulae can also procure us results which were not thought of when they were set up. Thus they can, for example, explain to us the phenomenon of aberration. A fixed star sends, for example, a light ray down along the Y-axis in system A. If the axis is positive in the ray's direction, one then has

$$h_x = 0, \qquad h_y = c, \qquad h_z = 0.$$

If B's system = the earth, one obtains from this, according to 10, for the latter

$$h'_x = -v, \qquad h'_y = \frac{c}{a}, \qquad h'_z = 0,$$

and thus, α being the aberration angle,

$$tg\,\alpha = -a\,\frac{v}{c},$$

a result which is only imperceptibly different from that found by the astronomers.

This calculation must however impress no one. For it is of course simply the usual way of deriving the aberration relation that is here repeated. Had we used two ordinary coordinate systems, we should have got the astronomical result itself out. Now we have used two somewhat artificial ones. Thereby the foreign constant a has come into the result. Fortunately it is so near 1 that it does not do much harm. But had it been widely different from 1, it would have come along all the same. One ought therefore surely to prefer to explain aberration in the way hitherto used, by the aid of two ordinary coordinate systems.

But the *Doppler* phenomenon too can be explained by the formulae! A light ray is sent from O along the X-axis in system A. If the state of vibration at O at the time 0 was $= U \cos 2\pi n\alpha$, it is at the time t at the place x

$$u = U \cos 2\pi n \left(t - \frac{x}{c} + \alpha \right).$$

But it will thus at the time t' at the place x' in B's system be

$$\begin{aligned} u' &= U\cos 2\pi na\left(t' + \frac{v}{c^2}\,x' - \frac{x'+vt'}{c} + \frac{\alpha}{a}\right) \\ &= U\cos 2\pi na\left(t'\left(1-\frac{v}{c}\right) - \frac{x'}{c}\left(1-\frac{v}{c}\right) + \frac{\alpha}{a}\right) \\ &= U\cos 2\pi na\,\frac{c-v}{c}\left(t' - \frac{x'}{c} + \beta\right). \end{aligned}$$

The vibration number has thus for B, who moves away from the light source with velocity v, become $n_1 = a\,\frac{c-v}{c}\cdot n$, thus very nearly as "*Doppler*'s principle" demands.

Here, however, there is if possible even less ground for satisfaction than in the previous case. For one can of course here maintain that the result is decidedly incorrect, and precisely as incorrect as one had to expect it on the basis of the two coordinate systems which, so to say, shrink under each other's gaze. By the aid of ordinary coordinate systems the *Dopplerian* result is obtained extremely easily, for example thus: if the velocity of light is $c = n\lambda$, there will pass a fixed point on the ray's path n waves in unit time; but if the point moves with velocity v away from the light source, the velocity of light in relation to the point becomes only $c - v$, and there will then pass it only $n_1 = \frac{c-v}{c}\cdot n$ waves in unit time. The original vibration number is thus multiplied by $\frac{c-v}{c}$, while the magnitude a has nothing whatever to do with the purely rational phenomenon.

That the theory of relativity can thus procure us various fairly correct results follows simply already from this, that the two coordinate systems in reality resemble two ordinary coordinate systems a good deal, and that the formulae in question likewise, on account of the great velocity of light and the correspondingly slight value of the magnitudes $\frac{v^2}{c^2}$ and $\frac{v}{c^2}$, continue to be tolerably usable. If $c = \infty$, all the formulae would of course pass over into the usual ones.

Indeed, one must even expect that in certain cases one will be able to get still more sensational results by the use of the formulae. For these are contrived so as to be able to make the enigmatic contradiction in the *Michelson* phenomenon apparently disappear, and everywhere where they lead to such an apparent annihilation of a contradiction there is therefore a certain probability that we stand face to face with the same enigmatic cause which has also made itself felt in the *Michelson* experiments and those akin to them. One must however take good care not to see in such results proofs of the theory's correctness.

Of course it is not entirely excluded either that the formulae may also for once, by a chance hit, lead to a correct result. Thus it is not easy to decide how the matter really stands with the following case, which concerns the velocity of light in flowing water:

Water flows in the X-direction with the constant velocity v. It is thus at rest for system B. A light ray is sent in the same direction through it. Its velocity in resting water is $\frac{c}{n} = h'_x$, n being the water's refractive index. But in system A one then obtains, according to 10, to a close approximation.

$$h_x = \frac{\frac{c}{n} + v}{1 + \frac{v}{cn}} = \left(\frac{c}{n} + v\right)\left(1 - \frac{v}{cn}\right) = \frac{c}{n} + v - \frac{v}{n^2} - \frac{v^2}{cn} = \frac{c}{n} + \frac{n^2 - 1}{n^2}\,v,$$

thus precisely *Fizeau*'s and *Michelson*'s result.

The reason why one may not here without further ado conclude that one has before one a phenomenon akin to the *Michelson* interference phenomenon is this, that here there was of course no contradiction which was removed by the use of the formulae.

We shall cast yet a glance at a couple of the velocity formulae and the time formulae, in order to show what we must accept if we accept these formulae.

Some body or other has in system B the X'-velocity h'_x, and system B itself has the constant velocity v. According to all sound sense one should then expect that the body in system A would have to get the sum of the two velocities. But the formulae bid us assume that the velocity in system A becomes only

$$h_x = \frac{h'_x + v}{1 + \frac{v}{c^2}\,h'_x}.$$

Therewith, then, the usual addition of velocities, indeed the whole parallelogram theorem, is abolished. And here too there is no sense in consoling oneself with the thought that the difference from the usual is only exceedingly slight. For this difference is at the same time a difference between *the comprehensible* and *the incomprehensible*. No one will after all be satisfied with a theory which declares $2 + 2 = 4 + \varepsilon$, even if ε is an ever so small magnitude.

Nor are the remarkable consequences exhausted with so little. If h'_x and v are both merely a trifle less than c, the formulae make h_x less than c as well. For if one sets $h'_x = c - \mu$ and $v = c - \nu$, where μ and ν are both positive, while they may otherwise be thought as small as one pleases, one obtains

$$h_x = \frac{2c - \mu - \nu}{1 + \dfrac{(c-\nu)(c-\mu)}{c^2}} = \frac{2c - \mu - \nu}{1 + \dfrac{c^2 - (\mu+\nu)c + \mu\nu}{c^2}}$$
$$= c\,\frac{2c - \mu - \nu}{2c - \mu - \nu + \dfrac{\mu\nu}{c}}.$$

We are likewise compelled to assume that if $h'_x = c$, h_x also becomes $= c$, quite independently of how great or small v may be. For one then has

$$h_x = \frac{c+v}{1 + \dfrac{v}{c^2}\,c} = c\,\frac{c+v}{c+v}.$$

Indeed, if one wishes to bring the absurdity out still a little more strongly, one may for example try to solve the following problem: let in system A a point (for example a light signal) at the time 0 run from O out along the X-axis with velocity c, and let system B (which of course may itself be a light signal) likewise have its velocity $v = c$. The point must then according to all sound sense continue to coincide with O', or in system B get the velocity 0. But sound sense and the formulae are two widely different things. If one keeps to the formulae, one gets

$$h'_x = \frac{c-v}{1 - \dfrac{v}{c^2}\,c} = c\,\frac{c-v}{c-v} = c.$$

The time formulae too lead to the most remarkable results.

We think, for example, of two phenomena, F_1 and F_2, which by A must be referred to the times t_1 and t_2 and to the places on the X-axis x_1 and x_2. Let t_1, t_2 and x_1 be $= 0$ and x_2 positive. The two events are thus for A simultaneous. For B they are determined by the magnitudes t'_1, t'_2, x'_1, x'_2, of which t'_1 and x'_1 must likewise be $= 0$, while for x'_2 and t'_2 one obtains

$$x'_2 = a(x_2 - vt_2) = ax_2$$
$$t'_2 = a\left(t_2 - \frac{v}{c^2}\,x_2\right) = -\,a\,\frac{v}{c^2}\,x_2 = -\,\frac{v}{c^2}\,x'_2.$$

For B the two phenomena thus become, according to the formulae, not simultaneous. F_2 comes before F_1. Simultaneity is thus according to the formulae something relative. What is simultaneous for one is not so without further ado for another.

This, for a change, we understand perfectly well. For the simultaneous is here defined as that which presents itself in like clock readings, and clocks can as is well known both go and stand differently, just as one must also bear well in mind that it is one thing that two events are simultaneous, and another that A or B gets word of them simultaneously. Several relativists, however, believe they have here made a significant discovery concerning time's inmost essence.

One has therefore also here become apprehensive, and has with a certain anxiety asked: but if time is something so relative, might it not be that the causal relation too showed itself relative, and that the effect merely for the one came after, but for the other before the cause? It is really not easy to understand why one has become anxious precisely here; for when one has on so many other points made existence illogical, one step more could hardly make much difference. Let us however look at the question!

Let for A F_2 come a little later than F_1 and be, for example, an effect of F_1 at the place x_2. Can it not then happen that F_2 for B comes before F_1, and the effect thereby before the cause?

F_1 came of course for B as well as for A at the time 0, and for B $t_2' = a\left(t_2 - \frac{v}{c^2}\,x_2\right)$. If t_2' is to become negative, one must thus have

$$\frac{v}{c^2}\,x_2 > t_2 \quad \text{or} \quad \frac{v\,\frac{x_2}{t_2}}{c^2} > 1.$$

Now suppose that F_1 and F_2 are a light signal's departure from x_1 and arrival at x_2; then $\frac{x_2}{t_2}$ = its velocity = c, and the condition for F_2 coming for B before F_1 will then be fulfilled if v is greater than c.

At this consequence (as well as at a number of other similar ones) one has however become so alarmed that one has sought to secure oneself by adding to the two presuppositions already named the further dogma that the velocity of light c is altogether the greatest velocity to be found in the world. The more firmly one believes in this assertion, the more firmly will one thus be able to hope that the effect — in spite of the two presuppositions — will never come before the cause. Now the velocity of light is certainly a great velocity. But there are nevertheless those who have emphasised that if gravitation is at all (like every other phenomenon here in the world) a temporal phenomenon, then it must spread with a velocity many times greater than that of light.

Someone might here, in support of the doctrine, possibly remark that since we *necessarily* must rely on the principle of causality and thus assume a cause before every change, it can be directly concluded from this that c *must* be the greatest velocity in the world. This objection is however untenable. For the feared consequence proceeds of course solely from the two mutually conflicting presuppositions. But in assuming them one has broken with the principle of contradiction, and one cannot then afterwards appeal to the principle of causality, since these two principles are merely two expressions for one and the same thing. Here too the formulae have merely done what one had to expect of them: standing as they do in conflict with the principle of contradiction, they have also turned the principle of causality upside down. The mathematical part itself is, as has been said, in order.

On the other hand it is, as already remarked, something of an inconsistency that one is particularly eager to save the principle of causality from destruction; for the whole theory otherwise reveals but slight respect for the causal relation. With a rather peculiar example of this we may close these more special remarks.

Let in the resting system A a man set about travelling around with a certain velocity, v. Of what will then happen *Einstein* reports the following: the man will then himself become a system B, and his clock will therefore suddenly set about going more slowly than while he was at rest. When he has come home again and is at rest, the clock will resume its former rate; but it will now (if it was originally in order) show somewhat too little[*)]. Indeed, one has even extended the transformation to hold for the man himself as well. He will during his motion live a little more slowly than before, and he will therefore on his homecoming be a little younger than those of his former contemporaries who stayed at home. It is thus quite simply a kind of fountain of youth that one has here discovered. It merely has the defect of being somewhat weak in its action; for according to the formulae the man will, even if he travels constantly at express-train speed, take a good 10 000 years to gain 1 second. However, the quantitative interests us less here. It was the causal relation we wished to examine, and for that it will already be sufficient to consider the clock a little more closely. Quite of its own accord this can hardly have adjusted itself at the journey's beginning to go more slowly, and at its end adjusted itself anew to go as before. And it would be just as incredible to assume that the man himself should have adjusted his clock to go wrong as long as he travelled. That B's clocks come to go more slowly than A's we can, on the presupposition that his system really moves, understand comparatively easily. For he has of course himself adjusted them to it by using light signals on the presupposition that he was at rest. But the travelling man can hardly have "regulated" his clock during his express-train journey by the aid of such signals, on the presupposition that he was not travelling at all, or that the earth was travelling backwards under him precisely with the train's velocity. It therefore becomes

[*)] Lorentz, Einstein, Minkowski: Das Relativitätsprinzip, 1913, p. 38.

a completely causeless phenomenon that *Einstein* has here introduced into his theory, and moreover a phenomenon which the theory unfortunately in certain respects has use for, while it in reality does not follow at all from its presuppositions.

For *Lorentz* the matter would be quite another, just as his theory is on the whole widely different from *Einstein's*, although the latter's adherents are also fond of claiming him for their side. *Lorentz* assumes a real contraction under all motion and has thus a real physical cause to appeal to. With *Einstein* the contraction is merely a conjecture in A's consciousness, and that such a thing should be able to alter clocks one can hardly well believe.

VII.

If we pass from lengths, times and velocities over to accelerations, masses, forces and so forth, everything becomes doubly fantastic. The formulae's demands upon our willingness to believe are considerably increased. With full right *Lorentz* finds it extremely strange that all these wonderful views have found as rapid an acceptance as they really have. But the explanation of this phenomenon lies perhaps after all not so very far away. The time was in reality to no slight degree prepared and suited for a theory like the *Einstein*ian. For the characteristic thing about this theory is of course this, that it troubles itself so exceedingly little about causal connection, about understanding of the whole. If only we can get formulae set up "which fit", and which can therefore procure us information about new particulars, all is well. If there were a resting ether, it would be able to become a bad protest against the relativity that has been set up. *Einstein* is therefore not fond of this assumption, and declares resolutely: there is no use for an ether.[*)] That he at the same time attaches himself to the *Maxwell*ian theory of light, and thus assumes electromagnetic vibrations without assuming a dielectric medium which can make the vibrations in question tolerably comprehensible, does not trouble him. He has in his theory to no slight degree confused mathematics with physics, and the same tendency in reality makes itself felt in not a few of the physicists of the present day.

It was emphasised above that there is a decisive difference between the *Lorentz*ian and the *Einstein*ian theory. Since both are so often brought into quite close connection, it will perhaps be in place to state the difference a little more precisely.

Lorentz is not a relativist. He assumes a resting ether, and can already thereby define real rest as rest in relation to the ether. But still more sharply can he distinguish between rest and translation, in that he assumes a real contraction under translation. That one

[*)] Lorentz, Einstein, Minkowski: Das Relativitätsprinzip, p. 28.

cannot *always* in the real case measure the contraction is a circumstance which does not alter the matter. *Lorentz* stands so far at the same standpoint as *Newton*.

Einstein, on the contrary, finds the ether superfluous and rejects it. And he cannot of course with any meaning assume any real contraction, since his whole doctrine of relativity would thereby be laid in the grave. If there were contraction, there would of course be a difference between rest and uniform motion, even if one could not always determine it.

It has been urged: but *Lorentz* and *Einstein* have of course quite the same formulae in a great multitude of cases. To this it must again be answered that mathematics is not yet physics. The same mathematical expression may in two different theories quite well have widely different meaning and value. When *Einstein*, in the case of the traveller in the A-system, uses his B-formulae without further ado, he thereby, as has been said, introduces a physical witchcraft. His B-formulae are here quite unjustified, since the man has evidently not, like B, regulated his clock by light signals. When *Lorentz* reckons with the same formulae, there is on the other hand, in accordance with his presupposition of a real contraction, good sense in the matter. It therefore seems to me physically confusing when one so often mixes the two views together by introducing expressions like "the Lorentz-Einstein transformation" and the like, and I see no otherwise than that one must come to the result that the *Einstein*ian theory, in spite of all the ingenuity to which it undeniably testifies, must nevertheless in the end be stamped as a decidedly self-contradictory and thereby impossible view, while the *Lorentz*ian continues to stand as a possible explanation of the *Michelsonian* riddle. That this explanation is perhaps not the final or the best that is to be found has already been intimated. The *Lorentz*ian contraction too leads of course to various somewhat odd results: in the face of a velocity c or above, the moved body's volume will of course become 0 or negative. However, this consequence lies presumably so far out in the periphery that at all events, as a provisional working hypothesis, it will be most correct to use the doctrine.